

MLPerf(TM) Storage v3.0 - System Description

Submitter: Alluxio
Division: CLOSED
System name: aws_c5n18xlarge_i3en24xlarge_8c_10s_8a
Workload: Training - 3D U-Net (unet3d), 8 simulated accelerators (b200, 1/client), read_threads=24
Storage system: Alluxio Enterprise AI 3.9 (RC17) over AWS S3, via Alluxio-FUSE (POSIX)
Result: Accelerator Utilization 92.63% (5-run mean; bar >= 90%)
Cloud: AWS (us-west-1), single availability zone

1. Hardware configuration & node counts

Clients (host nodes): 8 x c5n.18xlarge

- 8 FUSE clients, 1 simulated accelerator each (8 total)
- CPU: 72 vCPU (Intel Xeon Platinum 8124M/8259CL) Memory: 192 GiB
- Storage: EBS gp3 (root) Network: 100 Gbps ENA

Storage nodes: 10 x i3en.24xlarge

- 10 Alluxio storage workers (NVMe page store, consistent-hash)
- CPU: 96 vCPU (Intel Xeon Platinum 8259CL) Memory: 768 GiB
- Storage: 8x 7.5 TB NVMe SSD -> RAID0 (~55 TB) page store Network: 100 Gbps ENA

2. System & network topology

Benchmark access is POSIX/file via Alluxio-FUSE; AWS S3 is the under-file-system (UFS). All nodes in one AWS us-west-1 availability zone over 50/100 Gbps ENA.

```
8x c5n.18xlarge  --Alluxio FUSE (POSIX, read)--> 10x i3en.24xlarge workers
(MPI, 1 accel ea)                               shared etcd + coordinator
                                                |
                                                AWS S3 (UFS)
```

3. Software stack & versions

- Alluxio Enterprise AI 3.9 (RC17) (distributed cache)
- Alluxio-FUSE (POSIX client)
- etcd 3.5.x (mount table + worker membership)
- Ubuntu 24.04 LTS
- OpenJDK 17
- mlpstorage 3.0.x (uv-managed) + DLIO 3.0.x
- Backing store: AWS S3 (UFS)

4. Non-default configuration options

- Client access: read-only (no client-side write cache / FDB)
- alluxio.dora.client.ufs.fallback.enabled=false; alluxio.client.list.status.from.ufs.enabled=false
- alluxio.user.fuse.enforce.direct.io.read=true; alluxio.user.fuse.random.access=true
- async prefetch: thread=512, part.length=4MB, max.part.number=8, file.length.threshold=4MB
- worker.network.netty.file.transfer=TRANSFER; frame.decoder.buffer.cumulator=COMPOSITE
- FUSE JVM: -Xms32g -Xmx32g -XX:MaxDirectMemorySize=32g
- Worker: write.cache.enabled=true + FoundationDB (worker-side; client is read-only)

5. Notes

Simulated accelerators (no GPU). Machine-readable system parameters are in the companion systemname.yaml. CLOSED workload parameters per mlpstorage; storage-system configuration is unconstrained in CLOSED. Dataset backing store: AWS S3.